Employee Training Record

Date of Training: View FileHold for sign off evidence

Note: Rows can be added or deleted as needed.

Name / Title of Training Requirement	Document Number[1]	Rev	Reason for Training	Type of Training (check one)	Certified as Trainer (check one)	Effectiveness Evaluation (check one)
Quality System Regulation (QSR) Training	N/A	N/A	☐ Initial Training and/or New Document ☐ Document Revision ☒ Re-training	☐ Document Originator ☒ Read and Acknowledge ☐ Interactive ☐ External[2]	☐ Yes[3] ☒ No	☒ Pass[4] ☐ Fail[4] ☐ N/A Initial / date: DP 21 Jun 2024

[1]If no controlled document, attach copy of training materials or agenda / summary to this record.
[2] Attach copy of training materials / syllabus / summary to this record.
[3]The Trainer or Supervisor is required to sign the record unless the person is the Document Originator.
[4]Attach associated effectiveness evaluation documentation to this record.

By signing below, I am indicating I understand the requirements of the above document(s), my associated role / responsibilities and have had the opportunity to ask questions.

Employee Number	Printed Name	Signature	Date
N/A	Brian McVerry	View FileHold for sign off evidence	View FileHold for sign off evidence

Trainer Signature (if applicable):______N/A______ (Print First & Last Name) ______N/A______ (Signature) Date:____N/A____

Training Effectiveness Check

QSR Training - 2024

Name: Brian McVerry Date: 06/21/2024

1. How often can the FDA inspect a medical device manufacturer?
   - A. Whenever they have a reason
   - B. Every 2 years
   - C. When there is a serious customer complaint
   - D. All of the above

2. True or False? Only production employees are required to follow the Quality Manual.
   - A. True
   - B. False

3. What is an interchangeable term for "GMPs"?
   - A. DMR
   - B. CFR
   - C. QSR
   - D. None of the above

4. The management representative for SILQ is?
   - A. Brian McVerry
   - B. Tavi Cruz-Uribe
   - C. Verne Sharma
   - D. Tom Downey

5. Complete the following sentence: Medical Device Classification identifies the level of regulatory control that is necessary to assure the ______________ & ____________________ of a medical device.
   A. Safety / Effectiveness
   B. Size / Shape
   C. Price / Tax
   D. None of the above

6. Complete the following sentence: US Device GMPs are contained in ____ CFR Part _____.
   A. 50 / 820
   B. 21 / 820
   C. 21 / 807
   D. 21 / 822

7. What are Good Manufacturing Practices?
   A. Keeping all manufacturing work areas clean.
   B. Regulations that describe the methods, equipment, facilities, and controls required for producing human and veterinary products, medical devices and processed food.
   C. A morning stretch with co-workers.
   D. FDA's suggested rules to follow.

8. Pick the best answer: What is the responsibility of the Management Representative?
   A. Responsible for creating organization chart
   B. Responsible for writing quality policy
   C. Responsible for overall quality system
   D. Responsible for providing coffee to all co-workers

9. True or False? Any patient injury or potential injury must be reported to the FDA and appropriate regulatory bodies.
   A. True
   B. False

10. If I have to make a correction on a document, what do I do?
   A. Use correction fluid
   B. Scribble it out with red ink.
   C. Draw a single line through the error, correct the error, and then initial and date next to the correction.
   D. Rip up the document and start over.

11. Where can you find SILQ's Quality Policy?
   A. On the Internet
   B. On a traveler form
   C. In the Quality Manual
   D. In the Quality Policy document
   E. Both C & D

12. Why do we conduct quality system training every year?
   A. To keep ourselves entertained.
   B. Our Employee Training SOP suggests it might be a good idea.
   C. The Quality Assurance department likes to conduct training.
   D. Quality System Regulation requires it.

13. What is the CFR?

A. Code of Federal Regulations

B. Certificate For Registry

C. Certificate Free Reign

D. Code For Returns

14. Complete the following sentence: Complaints must be _________ and _________:

A. Embellished and entertaining.

B. Short and concise.

C. Documented and tracked.

D. Read and memorized.

To be completed by QA (Passing result is 12 out of 14 correct)

Test Result (circle one): **PASS** **FAIL** 14/14 correct

Reviewed by: Diana Porter Date: 21 Jun 2024

Silq
Quality System Regulation (QSR) Training
2024

FDA History

- FDA dates back to 1906 with the passage of the Pure Food & Drug Act. It prohibited interstate commerce in adulterated and misbranded food & drugs
- Upton Sinclair's 'The Jungle' which described the horrible condition of the meat-packing industry was the final precipitating force behind both a meat inspection law and a comprehensive food & drug law

FDA History (continued)

- 1938 - The Federal Food, Drug, and Cosmetic Act
  - S.E. Massengil Company's *Elixir Sulfanilamide* drug was responsible for the deaths of more than 100 people. The drug & the deaths led to the passage of the 1938 Food, Drug, and Cosmetic Act (FD&C Act)
    - The drug's formulation hadn't been tested for toxicity. At the time, the food & drug laws did not require that safety studies be done on new drugs.
    - Selling toxic drugs was bad for business & could damage a company's reputation, but it was not illegal.
    - Because no tests had been done on the new drug, the chemist who created the elixir solution failed to note one characteristic: the ingredient of diethylene glycol, a chemical normally used as an anti-freeze, is a deadly poison.
    - FD& C Act increased FDA's authority to regulate drugs.

FDA History (continued)

- 1962 - Kefauver-Harris Drug Amendments
  - Derived from hearings held by Senator Estes Kefauver.
    - As with the 1938 act, a therapeutic disaster compelled passage of the new law; in this case the disaster was narrowly averted. Thalidomide was a sedative never approved in this country; it produced thousands of grossly deformed newborns outside of the U.S.
- 1976 - Medical Device Amendments
    - Dalkon Shield (intrauterine device): Thousands of women injured. At the time this device was not subject to rigorous clinical testing because it was not classified as a drug under the terms of the Federal Food, Drug, and Cosmetic Act
    - Congress responded to the Dalkon Shield fiasco by enacting the Medical Device Amendments of 1976 which authorized the FDA to review the safety and effectiveness of medical devices like the Dalkon Shield.

Medical Device Amendments (MDA-1976)

- Medical devices, previously only subject to adulteration and misbranding were now subject to:
  - premarket approval for life-supporting devices
  - classification
  - records and reporting requirements
  - establishment registration
  - investigational device exemptions

Safe Medical Devices Act (SMDA-1990)

- Created concept of “generic” devices – substantial equivalence
- Added MDR requirement (Medical Device Reporting)
- Authorized FDA to order device recalls
- Increased scope of auditing ability

Medical Devices Classification

- Classification identifies the level of regulatory control that is necessary to assure the safety and effectiveness of a medical device (FDC Act § 513 [360c]):
  - Class I - general controls alone are sufficient to assure reasonable assurance of safety and effectiveness (GMP's). Includes devices such as manual instruments, bandages, casting.
  - Class II - general controls alone are not sufficient to assure reasonable assurance of safety and effectiveness. Require special controls such as standards, post-market surveillance, patient registries, guidelines, etc. Typically require 510(k) submission for market. Examples: orthopedic implants, endoscopes, infusion pumps, patient data manipulation software.
  - Class III - general and special controls not sufficient to provide reasonable assurance of safety and effectiveness. Typically these are life sustaining or life supporting devices such as pacemakers, respirators, etc. Usually, a PMA or PDP is required. [PMA stands for Premarket Approval; PDP stands for Product Development Protocol]

Devices classified within 16 medical specialties

- 862 = Chemistry/Toxicology
- 864 = Hematology/Pathology
- 866 = Immunology/Microbiology
- 868 = Anesthesiology
- 870 = Cardiovascular
- 872 = Dental
- 874 = Ear, Nose & Throat
- 876 = Gastro/Urology
- 878 = General Plastic Surgery
- 880 = General Hospital
- 882 = Neurological
- 884 = Obstetrical/Gyn
- 886 = Ophthalmic
- 888 = Orthopedic
- 890 = Physical Medicine
- 892 = Radiology

Terminology

- Pre-amendment device is a device introduced in interstate commerce prior to May 28, 1976
- Predicate device is a device in commercial distribution to which another device claims substantial equivalence
  - Substantial equivalence means that the new device is at least as safe and effective as the predicate

CFR - Code of Federal Regulations

- How does it all work?
- Agencies are empowered by Congress
- Agencies create regulations (also termed administrative laws)
- Codified laws/regulations are published daily in the Federal Register and are updated yearly in the Code of Federal Regulations
- Code of Federal Regulations
  - 50 Titles, 150 Volumes
  - Title 21 has 9 volumes, contains regulations pertaining to Foods, Drugs, and Cosmetics
  - §800 Contains:
    - Classification scheme for medical devices
    - GMPs also known as QSR

What are GMPs?

What are GMP's (QSR's)?

- Good Manufacturing Practices (GMP's) are regulations that describe the methods, equipment, facilities, and controls required for producing:
  - Human and veterinary drug products (21 CFR 210-211)
  - Medical devices (21 CFR 820)
  - Processed food (21 CFR 100)

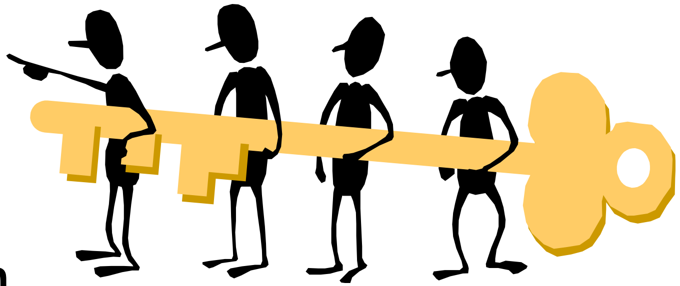
(QSR stands for Quality System Regulation)

Why do the GMP's (QSR's) exist?

- GMP's define a quality system that manufacturers use as they build quality INTO their products.

Why is so much paperwork required?

- To make sure we know exactly what we did, and when we did it.
- To be able to correct mistakes if they happen.
- To be able to PREVENT mistakes from happening in the future.



What's the price of product failure in our industry of medical devices?

- Answers may include death, reduction in the quality of life, or grieving families due to the loss of a loved one.
- We must maintain paperwork which provides a permanent record of what we did.

How were GMP's (QSR's) developed?

- Originally, GMPs were based upon the best practices of the industry.
- As technology and practices improved, the GMPs evolved as well.
- US Device GMPs are contained in 21 CFR Part 820

How to Think About Quality

- “Quality” means those features of products which meet CUSTOMER needs and thereby provide CUSTOMER satisfaction
- “Quality” means
  - freedom from deficiencies
  - freedom from errors that require doing work over again (rework) or that result in field failures, customer dissatisfaction, customer claims, and so on.
  - Higher quality usually “costs less”

Our Paperwork is a Product

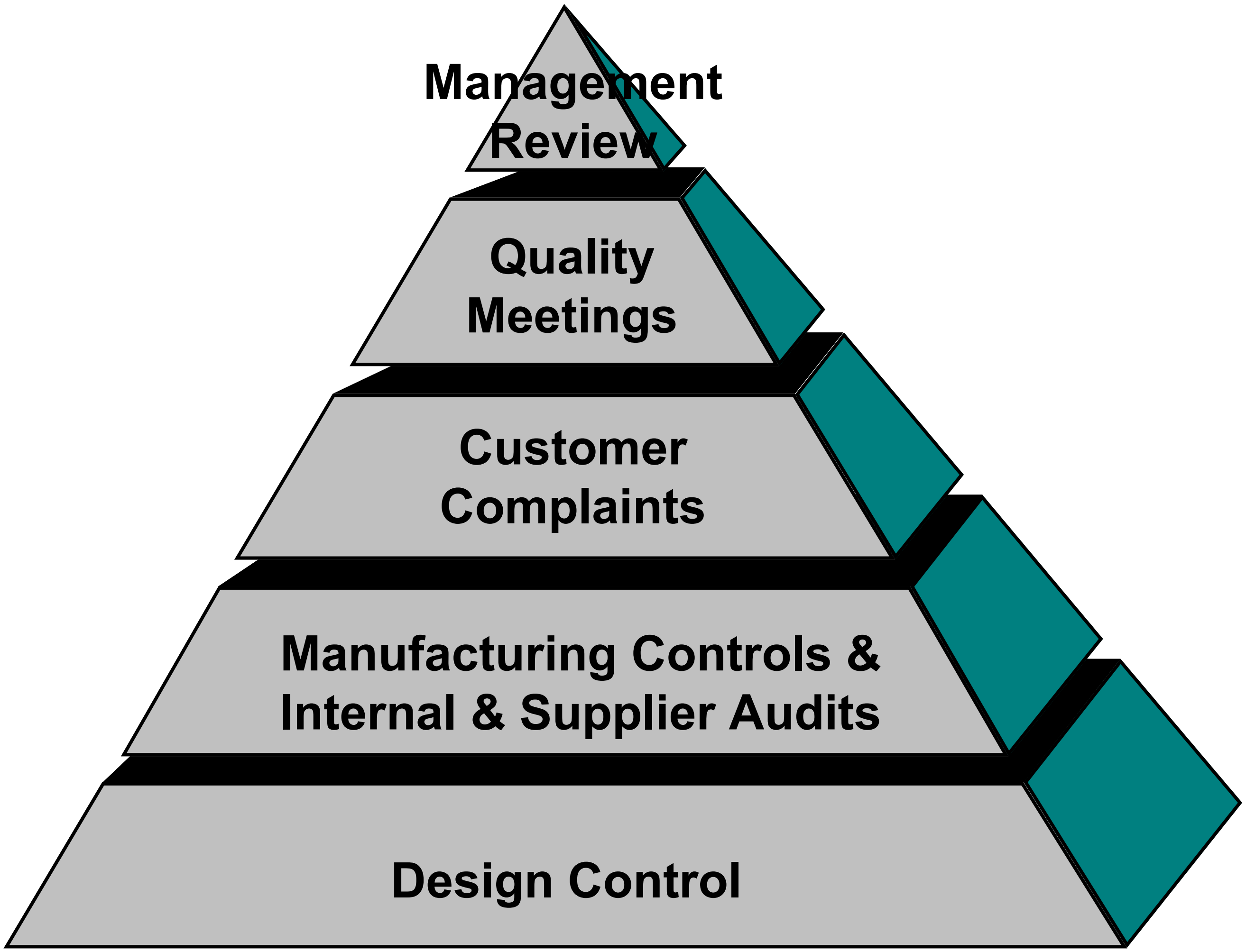
The paperwork we produce is of equal importance to the products we produce.

One of our customers is….FDA

FDA represents
our customers (hospitals,
patients, etc.).

They inspect our company to
ensure that our customers
can trust us.

Quality System



Remember this….

- SILQ'S Quality Policy can be found in the *Quality Manual* **and** *Quality Policy document*:

Excellent patient care is the focus of all SILQ's activities. We endeavor to meet or exceed our customers' satisfaction and expectations in providing safe and effective coated medical devices of the highest quality and that meets all applicable regulatory requirements.

Each SILQ employee is personally committed to achieving this goal and continually improving our ability to maintain customer focus by operating a highly effective quality system that meets the requirements of applicable global standards and regulatory requirements for medical devices.

And this….

- Management Representative
  - Responsible for overall quality system
  - Appointed by, and reports to, top management
  - Ensures the promotion of awareness of applicable regulatory and customer requirements throughout the company

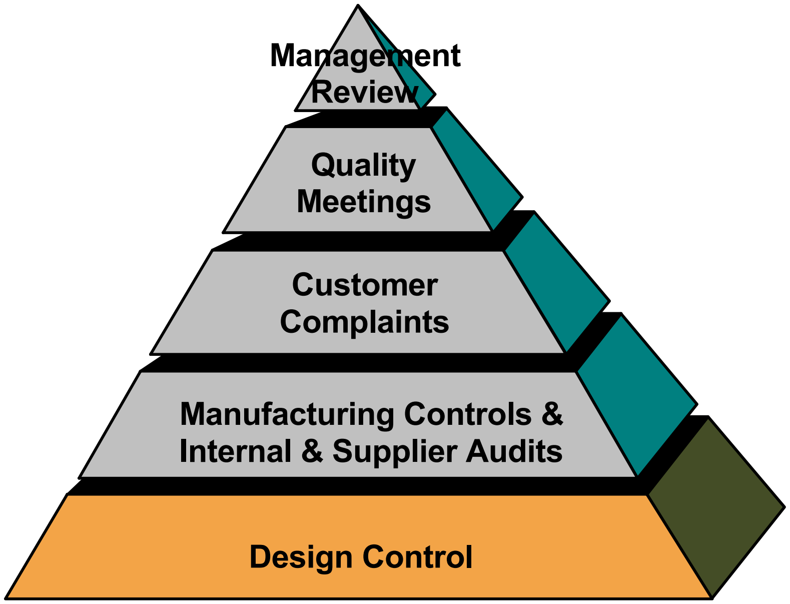
  - *Octavio Cruz-Uribe is SILQ's Management Representative*

Design Control:

Interrelated set of practices and procedures that are incorporated into the design & development process, i.e., a system of checks and balances. Design controls increase the likelihood that the design transferred to production will translate into a device that is appropriate for its intended use.

Elements of Design Control:

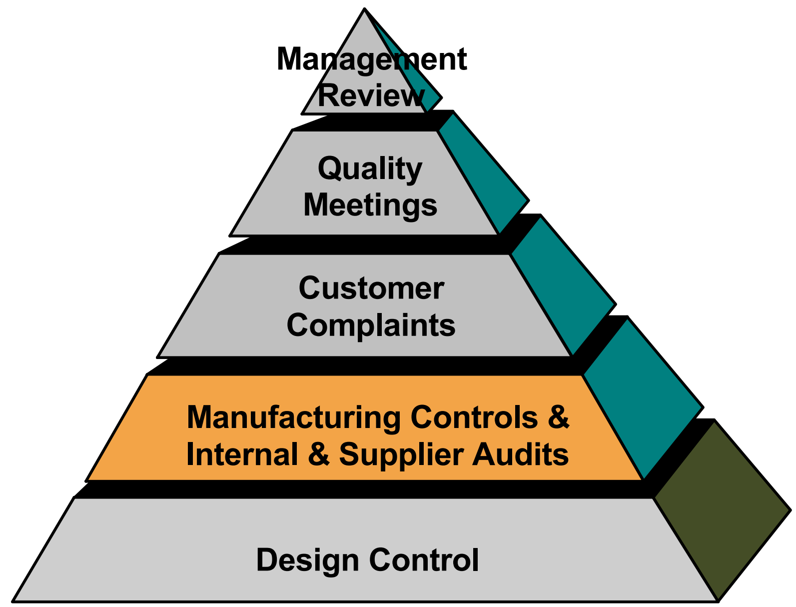
- Develop a design/development plan
- Design Input
- Design Output
- Design Review
- Design Verification
- Design Validation
- Design Transfer



Production and Process Controls

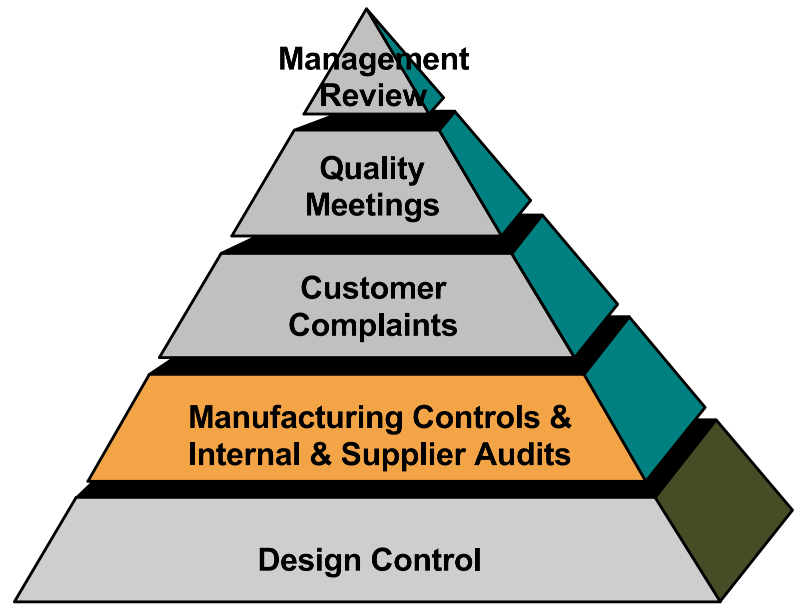
The purpose of this subsystem is to manufacture products that meet specifications.

- Procedures detailing
  - manufacturing steps
  - process controls
- Training
- Procedures for validating changes
- Equipment Calibration and Maintenance
- Validating Automated Processes
- Deviations
- Environmental Controls
  - Buildings
  - Personnel



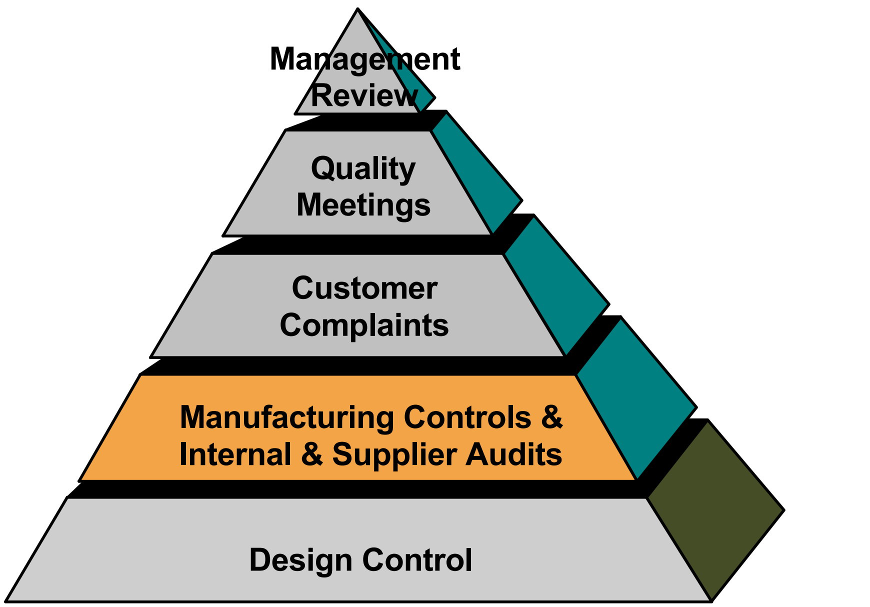
Production and Process Controls

- Packaging
- Identification & Traceability
  - Production Records
  - Labeling
- Acceptance Criteria
  - Final Inspection
  - Testing criteria
- Handling, Storage, & Installation



Production and Process Controls

- Nonconforming Product
  - identify, document
  - segregate
  - disposition (e.g., use as is, rework, return to vendor)
  - evaluation of nonconformance
- Purchasing Controls
- Supplier Selection and Audits
- Internal Audits



Complaint System

- Any Patient injury/potential injury must be reported to FDA and appropriate regulatory bodies
- Complaints must be documented and tracked
- Corrective actions
- Disposition of Product
- Recall

Audits

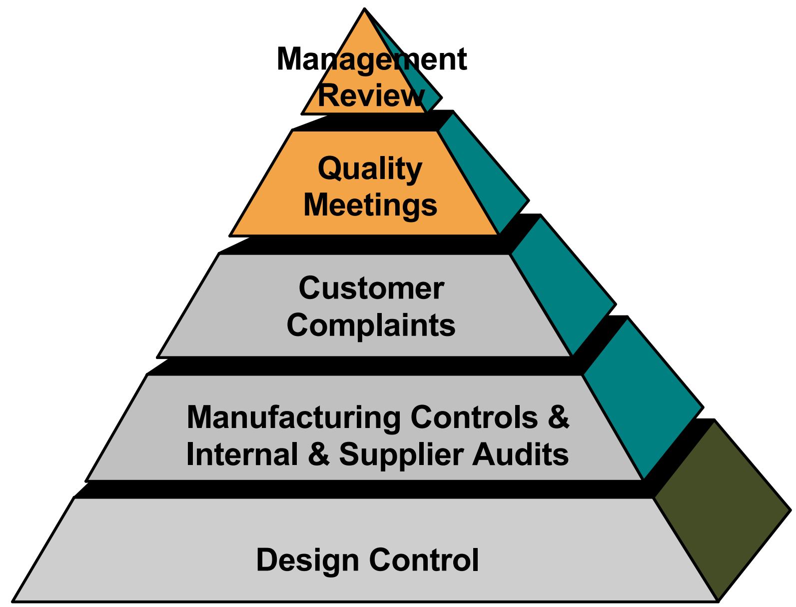
- Internal Audits
  - These are conducted every year to reveal how we measure up
  - And they show us where we can improve
- External Audits
  - Notified Body can perform announced and unannounced audits
  - FDA can inspect a medical device manufacturer at any time
- Supplier Audits answer the following questions:
  - How are our suppliers performing?
  - How can they help us make better products?

Corrective and Preventive Actions

- Problems happen: it's how you handle them that counts
- Track your progress and meet your timelines
- Sources of CAPAs include, but not limited to:
  - Clinical feedback and Complaints
  - Internal and External Audits
  - Nonconforming Materials
  - Management review
- "Correction" refers to repair, rework, or adjustment and relates to the disposition of an *existing* nonconformity.
- "Corrective Action" relates to the elimination of the *causes* of an existing nonconformity
- "Preventive Action" relates to the elimination of the causes of a *potential* nonconformity

Management Review Meetings

- Meetings held to review suitability & effectiveness of the quality system at defined intervals, with sufficient frequency, and according to established procedures.
  - Purpose:
    - Hold Senior Management Accountable
  - Items that we Review include, but not limited to:
    - Quality Policy and Objectives
    - Customer Feedback and Complaints
    - Internal / External Audit Results
    - Corrective and Preventive Actions
    - Nonconformances
    - Production Data



Document/Change Control

- We make sure to do things consistently, and with everyone "on the same page":
  - Standard Operating Procedures
  - Manufacturing Procedures and Travelers
  - User Manuals, Labeling
- All changes go through approval process (using DCO - Document Change Order).
  - This ensures that authorized approvers look over all proposed changes

Quality Records

- Rule Number One
  - If it isn't written down, (in the eyes of the FDA) it didn't happen.
- Rule Number Two
  - If it isn't written down properly, you're not positively sure exactly what happened.
- Rule Number Three
  - If the records aren't correct, neither is the product.

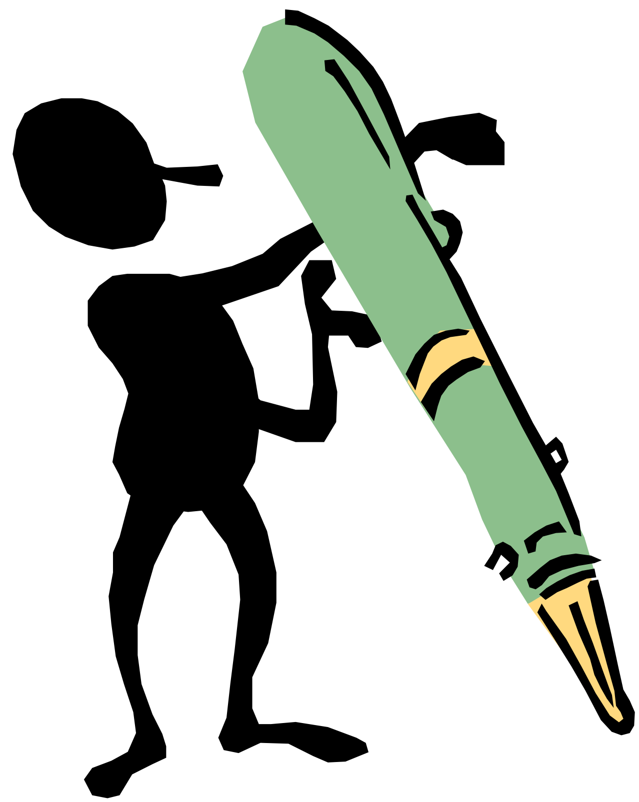
Quality Records

- Corrections made in ink (preferred colors: blue or black ink)
  - Draw a single line through the error
  - Make the correction next to the error
  - Initial and Date near the correction
  - No Post-it Notes, No White-Out
  - Do not write over a number or letter to correct it
    - Do <u>not</u> try to make a 5 into a 6 or a 6 into an 8!

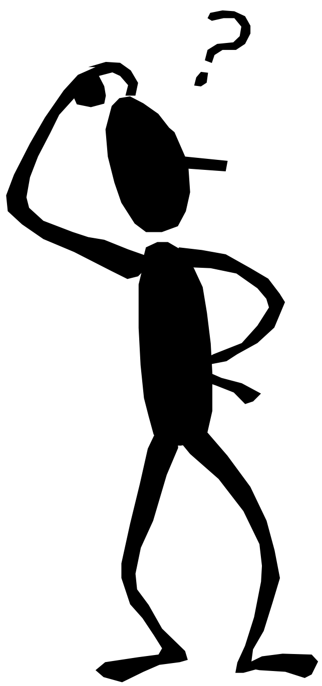
  - Don't leave blank spaces in forms and documentation
    - Mark through with a line or use "N/A"
    - Blank spaces need clarification so no one can come back & insert data "after the fact".
    - Because it appears that the form or document is incomplete and/or missing information

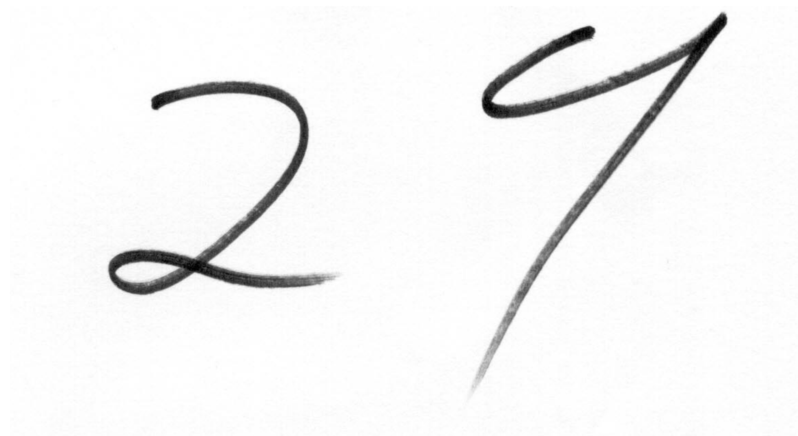
Record Keeping Tip #1

No pencils.



Record Keeping Tip #2



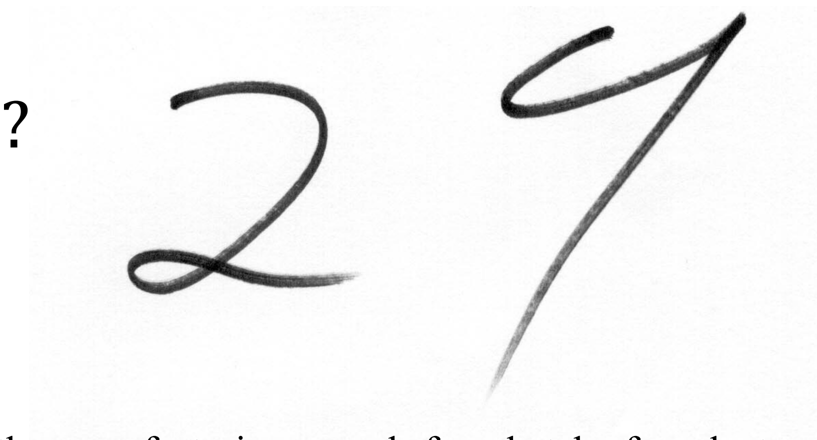


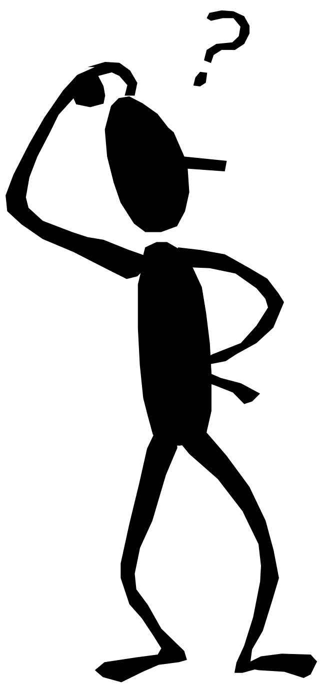
Ask yourself: Can everyone read what is written? No matter who, where, or what, can the written or printed data be read by everyone?

Record Keeping Tip #2

“What’s this number?”

Is it 29, 27, 24, or maybe even 2%?





Now suppose that you are reviewing the manufacturing records for a batch of product waiting to be shipped. There are 18 pallets of product already shrink-wrapped waiting to be loaded on a truck.

And let’s suppose that this value is a test result or some reading from a gauge. The specification is that 27 to 29 is acceptable.

So if the value is a 27 or a 29, then we’re in good shape. It meets our spec.

But what if it’s a 24? What then?

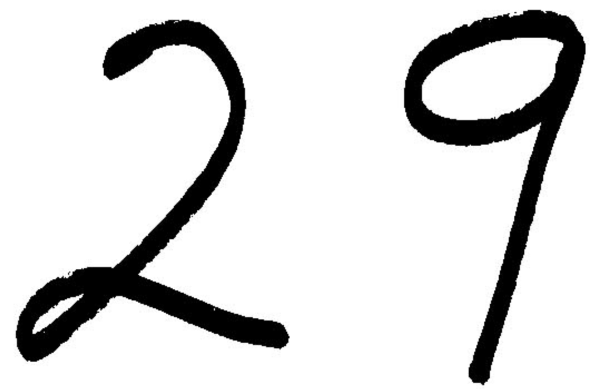
Do we scrap the lot? Can we take some samples and test for this again? If so, where do we get the samples?

Yes – break down the 18 pallets and obtain a representative sample. We might rush these samples into the lab, and ask for the test to be repeated – NOW! This puts additional pressure on the people working in the lab, further increasing the probability of contamination, mix-ups, and errors.

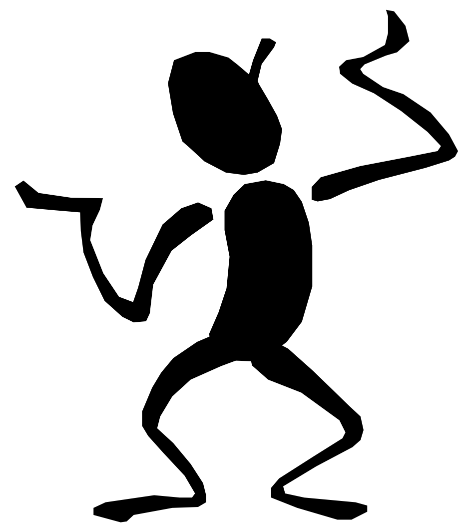
Record Keeping Tip #2

Sloppy writing is

100% preventable.



Take the time to write neatly!



Record Keeping Tip #3

No erasures.

No correction fluid.

No "Post-it" notes.

Defect Awareness

Each person should be aware of the importance of his or her individual contributions in the overall effort to achieve an acceptable level of quality.

All personnel should be aware that product quality is not solely the responsibility of management or any other single group.

Quality is the responsibility of every employee -- any employee can generate a quality problem through ignorance of their job requirements or negligence.

Quiz time!


DONE!